Introduction

Any hot body emits radiation. The energy radiated per second, is given by

$$Q = \sigma\,\epsilon\,A\,T^4 \tag{i}$$

where σ is the Stephan's Constant, ϵ is the emissivity of the body, A is the surface Area of the body and T is the temperature in Kelvin.

At the same time, the body also absorbs heat from the environment. The amount of heat absorbed per second is

$$Q = \sigma\,\epsilon\,A\,T_0^4 \tag{ii}$$

where T_0 is the room temperature. If heat is only lost through radiation, then the net rate of heat loss by the body is

$$Q = \sigma\,\epsilon\,A\,(T - T_0)^4 \tag{iii}$$

The Goal of this experiment is to determine the value of the Stephan's Constant σ.

Experiment Setup and Procedure

Apparatus

A constant Current source, DC differential amplifier, Stephan's Constant Box.

Experiment Setup

The Stephan's Constant box has a light aluminum cylinder suspended via a nylon wire. A heater is known resistance is connected to this. A thermocouple is attached that measures the temperature difference between the outer box and the aluminum cylinder. The EMF generated by the thermocouple is amplified using the DC differential amplifier. A constant current source is used to measure the pass a known current through the heater.

Procedure

The current was set at $196\;mA$ and the reading in the DC amplifier was allowed to stabilize. Once it stabilize, the value of voltage, current and room temperature were noted.

Next the current was reduced to $some\;mA$ and the process was repeated.

Formulae and Relations

Net heat emitted by the body (iii)

$$Q = \sigma\,\epsilon\,A\,(T^4 - T_0^4)$$

Power output by the heater:

$$Q = I^2 R$$

Temperature difference $= T - T_0 = V/\alpha$
Where V is potential across thermocouple and α is the thermo-electric power of thermocouple.

Data and Analysis

Room Temperature $T_0 = 300K$
We'll assume $\epsilon = 1$
Thermo-electric power of Thermo couple: $\alpha = 40\ \mu V/K$

Amplification factor of the DC differential amplifier $= 1000$

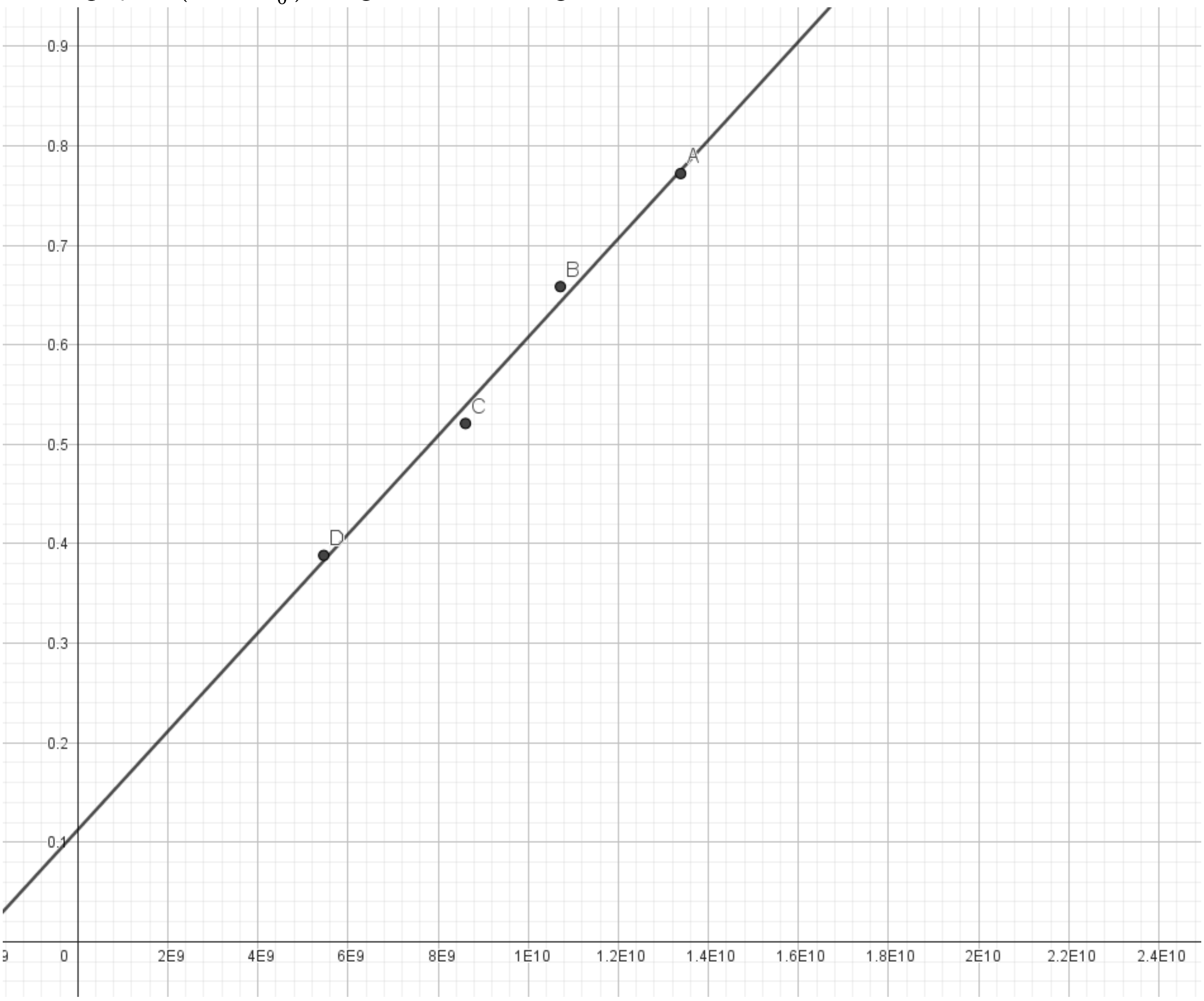
Resistance of heater: $R = 20.1\ \Omega$

Diameter of Aluminium Cylinder:
$d = 2\ cm = 2 \times 10^{-2}\ m$
$r = 1 \times 10^{-2}\ m$
Length of Aluminium Cylinder
$h = 3\ cm = 3 \times 10^{-2}\ m$
$\therefore A = 2\pi rh + 2\pi r^2 h = 2.513 \times 10^-3\ m^2$

The following reading were taken.

Current (mA)	**Voltage across Thermocouple (mV)**	**V (μV)**	$T = T_0 + V/\alpha$	$T^4 - T_0^4$	$Q = I^2 R$ (W)
196	3.314	3.314	382.85	13383973386	0.77
181	2.814	2.814	370.35	10712624884	0.66
161	2.381	2.381	359.52	8607688891	0.52
139	1.650	1.650	341.25	5460966408	0.39

Plotting Q vs $(T^4 - T_0^4)$ we get the following best fit line.



$y = 5 \times 10^{-11} x + 0.11$

The slope of the line = $\sigma \epsilon A$ = 5×10^{-11}

$\therefore \sigma = \frac{\text{slope}}{A} = 1.97 \times 10^{-8}\ W/m^2K^4$

Thus we have determined the value of Stephan's Constant.

The order of magnitude value obtained here is same as the established value of the Stephan's Constant. A main source of systematic error might be not waiting long enough for the voltage to stabilize, because as the system approaches equilibrium, the value of rate of decrease in voltage decrease. So, it is possible the reading was taken before the system and reached equilibrium and that will add to error. A more accurate method of determining the equilibrium point would be to analysis of how the temperature decreases with time and using that to determine what value the voltage is approaching.